Boao (James) Chen

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RESEARCH INTERESTS

Databases, Data Systems, Storage Engines, Machine Learning for Systems

EDUCATION

Doctor of Philosophy, Computer Science **Sep 2024 – Present**
Brandeis University, Waltham, MA
Advisor: Subhadeep Sarkar

Bachelor of Arts, Computer Science **Aug 2020 – May 2024**
Cornell University, Ithaca, NY
Washington University, St. Louis, MO

PUBLICATIONS

VLDB 2027 **Boao Chen**[*], Shubham Kaushik[*], and Subhadeep Sarkar. *Mind the Buffer: Dissecting the LSM-Buffer Design Space*, Proceedings of the VLDB Endowment, 20(3):784–797. [*]Equal contribution.

TPCTC 2025 Alexander H. Ott, Shubham Kaushik, **Boao Chen**, and Subhadeep Sarkar. *Tectonic: Bridging Synthetic and Real-World Workloads for Key-Value Benchmarking*, Performance Evaluation and Benchmarking TPC Technology Conference.

MobiCom 2024 Ke Li, Ruidong Zhang, **Boao Chen**, Siyuan Chen, Sicheng Yin, Saif Mahmud, Francois Guimbretiere, and Cheng Zhang. *GazeTrak: Exploring Acoustic-based Eye Tracking on a Glass Frame*, International Conference on Mobile Computing and Networking.

CHI 2024 Ke Li, Ruidong Zhang, Siyuan Chen, **Boao Chen**, Mose Sakashita, Francois Guimbretiere, and Cheng Zhang. *EyeEcho: Continuous Facial Expression Tracking on Glasses with Minimally-obtrusive Acoustic Sensing*, ACM Computer Human Interaction Conference.

IMWUT 2024 Ke Li, Devansh Agarwal, Ruidong Zhang, Vipin Gunda, Tianjun Mo, Saif Mahmud, **Boao Chen**, Francois Guimbretiere, and Cheng Zhang. *SonicID: User Identification on Smart Glasses with Acoustic Sensing*, The Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies.

POSTER

NEDB 2026 Shubham Kaushik, Steven Yang, **Boao Chen**, and Subhadeep Sarkar. *FluidLSM: A Fluid Log-Structured Merge Tree for Shifting Workloads*, North East Database Day.

NEDB 2026 Alexander Ott, Shubham Kaushik, **Boao Chen**, and Subhadeep Sarkar. *Tectonic: Bridging Synthetic and Real-World Workloads for Key-Value Benchmarking*, North East Database Day.

NEDB 2025 Alexander Ott, **Boao Chen**, Arpita Saha, and Subhadeep Sarkar. *Self-Designing LSM Memory Buffer*, North East Database Day.

TECHNICAL SKILLS

Programming Languages: C, C++, Java, Python, SQL, JavaScript, Node.js, R, MATLAB
Databases: RocksDB, MariaDB, Postgres, Cassandra, MongoDB
Machine Learning: PyTorch, TensorFlow, keras

RESEARCH PROJECTS

- **Mind the Buffer** (*VLDB 2027*): Conducted a comprehensive analysis of the LSM-tree in-memory buffer design space—the critical component through which every read and write operation must pass—evaluating how buffer implementation, sizing, and tuning bottleneck NoSQL storage engine performance across shifting workloads and hardware. Benchmarked 11 distinct buffer designs—including three hash-hybrid variants—and distilled the empirical tradeoffs into a practitioner handbook of 10 actionable guidelines for buffer selection and tuning. Designed and evaluated an adaptive buffer that dynamically switches implementations on-the-fly to sustain peak performance across workload shifts.

TEACHING ASSISTANT

Spring 2026	Introduction to Problem Solving in Python (COSI 10a), Brandeis University
Fall 2025	Capstone Project for Software Engineering (COSI 166B), Brandeis University
Spring 2025	Database Management Systems (COSI 127B), Brandeis University
Fall 2024	Algorithms (COSI 180A), Brandeis University
Spring 2024	Applied High-Performance and Parallel Computing (CS 5220), Cornell University
Fall 2023	Foundations of Artificial Intelligence (CS 4700), Cornell University
Spring 2023	Operating Systems (CS 4410), Cornell University

PROFESSIONAL SERVICES

2026	Student Volunteer for VLDB 2026
2026	Web Chair for Northeast Database (NEDB) Day 2026
2026 – Present	Member, Institute of Electrical and Electronics Engineers (IEEE)
2025 – Present	Member, Association for Computing Machinery (ACM)

CERTIFICATION

Jun 2025	*Oracle Certified Professional, MySQL 8.0 Database Administrator* – [link]
Jun 2025	*Oracle Certified Associate, MySQL Implementation* – [link]